

CHARCUBE · CALCULATION NOTE

CharCube sizing and first-principles checks

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DOCUMENT CONTROL

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REVISION HISTORY

Version	Date	Author	Change
0.5	2026-10-01	Amish Chadha	Budget treated as a value-engineering target
0.4	2026-09-30	Amish Chadha	Re-run for the constructable design (CCB-DDR-003); masses, tripod legs on fins, flue length; no requirement status changed
0.3	2026-09-26	Amish Chadha	Budget top-up approved by Amish
0.2	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002); baffle insert, jacket and lid blankets, 30 throat air holes, R5 at 5 h, R9 to 700 °C; all tables re-run
0.1	2026-09-25	Amish Chadha	First TRL 3 sizing note (batch, char, heating time, shell loss, masses, energy balance, draft and air, jacket, cooling, carbon, logger, cost) against every requirement

CharCube sizing and first-principles checks

On paper the nested-drum retort works as a kiln, and with the changes Amish accepted on 2026-09-25 (CCB-DDR-002) it now also works as a water heater. Six of the twelve requirements are met, five are at risk and one cannot be verified at TRL 3. R10 is within the \$320 value-engineering target: the spiral baffle insert, jacket blanket and lid blanket take the estimated cost of the kiln parts to \$319, \$1 under the target. The baffle and jacket blanket raise the heat into the water from 6.4 MJ to about 11.5 MJ (R6 at risk), the lid blanket cuts the wood from 5.8 kg to about 3.7 kg (R8 at risk), and R5 was relaxed to 5 h (at risk). The 25 mm secondary air ring of TRL 2 would have starved the burner throat; the model uses an air shroud with 30 air holes.

Every number in this note is printed by `docs/04-calcs/sizing.py` (run from the repo root: `python docs/04-calcs/sizing.py`), which also writes `docs/04-calcs/results.csv`. The script reads the geometry from `PARAMS` in `cad/src/model.py` and the prices from `bom/bom.csv`, so the model, the drawing CCB-DWG-001 and this note agree. All values are first-principles estimates; nothing here is measured.

■ 1. Assumptions

Table 1. Inputs. All are assumptions for a paper design unless marked as decided.

Input	Value	Basis
Kiln, jacket, throat, logger, blanket, tripod	As decided	CCB-DDR-001 items 1 to 8
Air shroud, baffle insert, jacket and lid blankets	As decided	CCB-DDR-002 items 11 to 13
Feedstock	Bundled straw, maize or cotton stalks and cobs, packed to 120 kg/m ³ air-dry at 12 % moisture (wet basis)	Decided feedstock, CCB-DDR-001 item 4; density as TRL 2
Feed, dry ultimate analysis	C 45, H 5.8, O 40.4, N 0.8, ash 8.0 %	Typical for maize and cotton stalks; straw-rich loads have more ash
Char	C 55, H 2.5, N 1.0 %, ash from a mass balance, O by difference	Yield 28 % of dry feed (20 to 35 %), as TRL 2
Heating values	Channiwala and Parikh correlation from the analyses	Channiwala and Parikh 2002, Fuel 81: 1051 to 1063
Start-up wood	Air-dry, 12 % moisture; C 50, H 6.0, O 43.2 % dry	Prunings or split wood
Burn	Annulus gas held at 650 °C, retort wall at 600 °C, until the core reaches 450 °C, then 30 min more (R3)	Operating intent
Charge	Effective heat capacity 2.72 kJ/kg K per kg dry (sensible, drying and 0.4 MJ/kg net pyrolysis heat); bed conductivity 0.20, 0.35 or 0.50 W/m K	Packed straw is about 0.1 W/m K cold; radiation in the pores raises it when hot
Air	Overall excess air ratio 1.6, 2.0 or 2.4; 40 % of the air enters as secondary air at the throat; 15 % of the volatiles burn in the throat	Small batch kilns run lean; damper setting
Combustion efficiency	95, 90 or 85 % of the lower heating value	Allows for CO, methane, tar and soot
Blankets	25 mm ceramic fibre, 128 kg/m ³ , 0.09 W/m K at a mean of about 350 °C, on the side and on the lid and top band; 25 mm mineral wool, 0.05 W/m K, on the jacket shell	Supplier data typical; air ports, throat and the lid under the shroud bare
Spiral baffle insert	Gas-side convection in the sleeve multiplied by 3; 4 extra velocity heads of pressure loss	Assumption for a twisted strip in laminar and transitional flow; to be measured (TRL 4, on hold)
Outside surfaces	Emissivity 0.9, still air at 20 °C	Worst case for surface temperature is sun and no wind
Carbon	100-year permanence factor 0.80 (IPCC medium temperature); methane 24 g per kg of char at GWP100 28	As CCB-PRC-001 v0.2, with sources there

Three scenarios bracket the result: **favourable** (bed conductivity 0.50 W/m K, excess air 1.6, efficiency 95 %, yield 35 %), **central** (0.35, 2.0, 90 %, 28 %) and **unfavourable** (0.20, 2.4, 85 %, 20 %). Central values are quoted unless a range is given.

■ 2. Batch and char (R1, R2)

The retort holds 100 L of packed feed at a 600 mm fill depth (123 L geometric volume), which is **12.0 kg** air-dry: 10.58 kg of dry matter and 1.44 kg of water. R1 is **met** for the decided feedstock. Loose chopped straw at 40 kg/m³ would give only 4.0 kg; it is outside the decided feedstock (CCB-DDR-001 item 4).

At 28 % yield the batch makes **2.96 kg** of char (2.12 to 3.70 kg). All the feed ash stays in the char, so the char is about 29 % ash (23 to 40 %), and its heating value is about 20.2 MJ/kg, lower than the 24 MJ/kg assumed at TRL 2. About 34 % of the feed carbon stays in the char. R2 is **at risk**: the 28 % central estimate clears the 25 % target, but the range does not.

A straw-rich load with 15 % ash would push char ash above 45 %, which is not consistent with 55 % carbon; the script warns when that happens.

■ 3. Heating time (R3, R5)

The burn is set by heat conduction into the packed charge, not by the fire. A one-term solution for a finite cylinder (radius 0.23 m, 0.6 m high, heated on all faces) gives the core 450 °C after:

Table 2. Heating and burn time.

Scenario	Bed conductivity	Core at 450 °C after lighting	Light to end of burn (R5)
Favourable	0.50 W/m K	2.9 h	3.4 h
Central	0.35 W/m K	4.0 h	4.5 h
Unfavourable	0.20 W/m K	6.5 h	7.0 h

The times include 0.5 h of warm-up and a 30 min hold at 450 °C for R3. R3 is **at risk**: it is reachable if the annulus is held hot long enough, which is what drives the wood demand in section 5. R5, relaxed to 5 h by Amish's decision (CCB-DDR-002 item 14), is met on the central estimate but **at risk** because the unfavourable case takes 7.0 h. The blankets do not change the burn time, which conduction into the charge sets. Sealed cooling (section 8) is short enough for the 16 h unload limit.

■ 4. Shell loss, surfaces and masses (R11, R12)

Table 3. Heat loss during the burn (annulus gas 650 °C).

Surface	Area	Surface temperature	Loss
Blanketed side	1.33 m ²	125 °C	2.20 kW
Port band, bare	0.25 m ²	168 °C	0.69 kW
Lid and top band, blanketed	0.28 m ²	120 °C	0.43 kW
Lid under the air shroud, bare	0.04 m ²	257 °C	0.24 kW
Burner throat, bare	0.20 m ²	223 °C	0.91 kW
Total			4.47 kW

Without the side blanket the side would run at about 311 °C and lose 11.3 kW, so the side blanket (decided, item 7) roughly halves the loss. The lid and top band blanket (CCB-DDR-002 item 13) cuts that part from 1.90 kW (257 °C, v0.1) to 0.67 kW, a saving of **1.22 kW**, a little less than the 1.4 kW estimated in v0.1 because the lid under the shroud stays bare so the secondary air is not blocked.

Table 4. Masses.

Part	Mass
Outer drum with hoops and ring	18.8 kg
Outer lid with collar	3.4 kg
Retort with lid and U-bolt handles	12.6 kg
Burner throat; air shroud with band damper	3.2 kg; 2.0 kg
Firebricks (3); side blanket; lid blanket	2.6 kg; 4.3 kg; 1.0 kg
Water jacket, empty, with fins and loose lid	23.3 kg
Spiral baffle insert; jacket shell blanket	1.2 kg; 2.2 kg
Flue with clips and cap; tripod legs, cleats and pads; tap	6.4 kg; 13.7 kg; 0.6 kg
Kiln without plinth and water	97 kg
Heat-recovery lift unit (tripod, drained jacket with baffle and blanket, flue, tap)	47.4 kg; 23.7 kg each for two people
Retort with char	15.5 kg; one person
Jacket full (61.4 L at 540 mm depth)	85 kg, never moved full

R11 is **met**: no lift exceeds 25 kg per person, but only if the jacket is drained and two people lift the unit, and the margin is now 1.3 kg per person. The only welded part is the jacket, with its three leg fins; the collar, shroud, baffle and tripod are riveted, bolted or folded (CCB-DDR-003). Each tripod leg (40 x 40 x 4 mm angle bolted to a jacket fin and pinned at a foot cleat, 1.43 m between the foot pin and the lower fin bolt, slenderness 184) carries about 327 N with a full jacket against an Euler load of 18.9 kN, a factor of about 58.

R12 is **met**: the flue outlet is 2.70 m above the ground and no BOM line calls for galvanized parts. The 5 m clearance is an operating rule (CCB-PRC-001, Safety).

■ 5. Energy balance and start-up wood (R8)

Wood makes up whatever the burning volatiles cannot supply while the annulus is held at 650 °C for the whole burn. The sinks are the retort charge and steel (19 MJ), the stored heat in the drum, lid, throat, bricks and blankets (9 MJ), the shell loss over the burn, and the heat carried out by the annulus gas.

Table 5. Energy per batch.

Quantity	Favourable	Central	Unfavourable
Burn time	3.4 h	4.5 h	7.0 h
Volatiles, lower heating value	104 MJ	117 MJ	132 MJ
Energy left in the char	73 MJ	60 MJ	45 MJ
Shell loss over the burn	55 MJ	72 MJ	113 MJ
Wood needed (R8)	1.7 kg	3.7 kg	9.8 kg
Heat released in the kiln	125 MJ	160 MJ	247 MJ
Flue gas per batch	79 kg	128 kg	246 kg
Gas leaving the throat (jacket inlet)	471 °C	432 °C	396 °C

Light-up alone needs about 1.7 kg of wood; in the favourable case that is all the wood needed. R8 is met on the central estimate (3.7 kg against 5 kg) but **at risk**, because a long burn with a poorly conducting charge needs about 9.8 kg. Excess air matters as much as burn time: every kilogram of surplus air leaves at 650 °C. Figure 2 in CCB-PRC-001 shows the central energy flow: 253 MJ in (residue at its higher heating value, wood at its lower), 60 MJ kept in the char, 33 MJ as latent heat and unburned gas, 160 MJ released, 99 MJ to the charge, structure and shell, and 61 MJ in the flue gas at the jacket, of which 11.5 MJ goes into the water.

■ 6. Gas flow, draft and air (R4)

Table 6. Flow checks at the central case; peak flow is taken as twice the mean.

Check	Result
Volatiles leaving the retort, mean and peak	0.73 and 1.46 g/s
Eight 20 mm gas holes (25.1 cm ²) at peak	1.3 m/s, 1.0 Pa: little back-pressure in the retort
Flue gas, mean and peak	8.0 and 15.9 g/s
Draft, air ports to outlet (2.43 m)	16.7 Pa
Losses at peak: flue path (including 4 velocity heads for the baffle) and primary ports	5.6 and 0.17 Pa; margin 2.4
Throat velocity at peak; flue velocity above the jacket	1.69 m/s; 0.75 m/s
Secondary air at peak (40 % of air)	4.88 g/s
Suction at the throat air holes at peak	5.2 Pa
Secondary hole area needed and provided	28.6 and 33.9 cm ² (30 x 12 mm in two rings, 34 mm pitch)
TRL 2 25 mm inlet pipe at the same flow	7.2 m/s and about 46 Pa: would not work
Air shroud intake (230 mm sleeve, 204 cm ²)	0.20 m/s

The draft is enough, with a margin of about 2.4 at peak (3.2 before the baffle). The baffle's resistance lowers the suction at the throat holes from 7.6 to 5.2 Pa, so the 24 holes of v0.1 (27.1 cm²) would have been about 5 % short; the model now has 30 holes with about 19 % spare area. The air shroud was accepted by Amish (CCB-DDR-002 item 11). R4 is **not verifiable at TRL 3**: the gas route is closed by design and the air paths are sized, but smoke and methane depend on mixing and temperature and need measurement.

■ 7. Water jacket (R6, R7)

The 600 mm flue sleeve (164 mm bore, 0.31 m² wetted) sees gas at 396 to 471 °C and a Reynolds number of about 1,400 to 2,100, which is laminar or transitional. A plain sleeve gives only about 1 to 3 W/m²K by convection and 2 to 2.5 W/m²K by gas radiation. The spiral baffle insert (CCB-DDR-002 item 12) is taken to triple the convection, to about 4 to 7 W/m²K, so the sleeve takes about 21 to 24 % of the gas heat, and the mineral wool blanket keeps most of it in the water.

Table 7. Heat into the water per batch (61.4 L, starting at 20 °C).

Case	Heat to water	Water temperature rise
Favourable (short burn)	9.0 MJ	
Central, design (baffle and jacket blanket)	11.5 MJ	about 45 K
Unfavourable (long burn)	19.1 MJ	
Central, plain sleeve and bare jacket (v0.1 design)	5.6 MJ	
Central, jacket blanket only	6.7 MJ	
Central, baffle insert only	9.5 MJ	

The comparison rows use the v0.2 flue gas flow, which is lower than in v0.1 because less wood is burned; that is why the plain sleeve now gives 5.6 MJ, not 6.4 MJ. R6 is **at risk**: the central case meets 10 MJ, but a short burn gives about 9.0 MJ, and the factor of 3 for the insert is an assumption until measured. The water does not boil in any scenario (boiling needs 20.6 MJ), but the long burn reaches 19.1 MJ, so R7 (open vent, loose lid, no sealing valve) is essential. R7 is **met** by design.

The sleeve wall and the baffle run close to the water temperature, far below the tar dew point, so any tar that escapes the throat will deposit there and cut heat transfer. The baffle hangs from the foot of the flue and lifts out with it for cleaning. While the water is below about 41 °C, water vapor from the flue gas condenses on the sleeve and will drip back into the throat.

■ 8. Cooling and unloading (R5)

With the ports and damper closed, the kiln and char (30 kJ/K) lose heat through a shell conductance of about 10.2 W/K (lower than v0.1 because the lid is blanketed), a time constant of about 0.8 h, so the shell falls to 60 °C in about 2.1 h. The char core then needs about 1.6 h more by conduction. The sum, about **3.6 h**, puts unloading at about 8 h after lighting in the central case, inside the 16 h limit. The char should still be left sealed overnight, because air reaching hot char can reignite it (CCB-PRC-001, Safety).

■ 9. Carbon

Table 8. Carbon per batch (100-year basis).

Scenario	Carbon stored	CO ₂	Methane penalty	Net
Favourable	1.63 kg C	6.0 kg	2.5 kg CO ₂ e	3.5 kg CO ₂ e
Central	1.30 kg C	4.8 kg	2.0 kg CO ₂ e	2.8 kg CO₂e
Unfavourable	0.93 kg C	3.4 kg	1.4 kg CO ₂ e	2.0 kg CO ₂ e

At 250 batches a year the central case makes about 740 kg of char and a net removal of about **0.70 t CO₂e**, and keeps about 3.0 t of residue out of open burning. The methane penalty assumes the retort field average; a working burner throat would lower it, but that cannot be shown without measurement. Start-up wood is assumed to come from residues or prunings and is not counted as an emission.

■ 10. Logger (R9)

A type K class 2 probe is good to ± 5.2 °C at 700 °C; with the MAX31855 amplifier (± 2 °C from -200 to 700 °C, [datasheet](#)) the combined error is about ± 5.6 °C, just outside ± 5 °C. Class 1 (special limits) probes give about ± 3.4 °C to 700 °C. R9 was restated by Amish's decision (CCB-DDR-002 item 15) as ± 5 °C from 0 to 700 °C and indicative above, so it is **met** with class 1 probes. A 0.25 W logger uses about 6 Wh in 24 h against about 31 Wh usable from a 10,000 mAh power bank, and writes about 276 kB a day.

■ 11. Cost (R10)

Value-engineering target: \$320 (a hypothetical control target, not a limit). The 17 BOM lines give an estimated cost of **\$319**, \$1 under the target. Lines 15 to 17 (baffle insert \$7, jacket blanket \$8, lid blanket \$6; \$21 in all) were added by Amish's decisions on items 12 and 13 (CCB-DDR-002). The safety kit (about \$40) is required and listed separately in `bom/bom-notes.md`, outside the kiln parts target (CCB-DDR-001 item 2). The script reads the target from `project.yaml`. R10 is **within the value-engineering target**, by only \$1.

■ 12. Results against requirements

Table 9. Every requirement in CCB-REQ-001 v0.4. Also written to `docs/04-calcs/results.csv`.

ID	Requirement (short)	Value (central, range)	Target	Status
R2	Char yield	28 % (20 to 35 %); 2.96 kg	25 % or more	At risk
R3	Core 450 °C for 30 min	Core at 450 °C 4.0 h after lighting (2.9 to 6.5 h) if the annulus is held at 650 °C	450 °C, 30 min	At risk
R5	Light to end of flaming; unload	Burn 4.5 h (3.4 to 7.0 h); unload after about 8 h	5 h or less (relaxed); 16 h or less	At risk
R6	Heat into water per batch	11.5 MJ (9.0 to 19.1 MJ) with baffle and jacket blanket	10 MJ or more	At risk
R8	Start-up wood per batch	3.7 kg (1.7 to 9.8 kg); light-up alone 1.7 kg	5 kg or less	At risk
R4	Burn the gas; smoke; methane	Route closed by design; draft margin 2.4; air holes sized	Methane below 24 g/kg	Not verifiable at TRL 3
R10	Kiln parts cost	\$319 (lines 15 to 17 add \$21)	\$320 or less (target)	Within the value-engineering target
R1	Batch size, decided feedstock	12.0 kg packed (loose straw 4.0 kg, excluded)	10 kg or more	Met
R7	Open water circuit	Open vent, loose lid, tap at base, tripod-carried	By design	Met
R9	Logger accuracy and endurance	±3.4 °C to 700 °C with class 1 probes; 6 of 31 Wh	±5 °C to 700 °C (restated)	Met
R11	Crew and lifts	23.7 kg each for two; retort with char 15.5 kg	25 kg per person or less	Met
R12	Outlet height; no galvanized hot parts	2.70 m; none	2.5 m or more	Met

■ 13. Limits of this note

- The heating-time model treats the charge as a solid with a fixed effective conductivity. Real beds shrink, crack and release gas, which can speed or slow heating; the three scenarios bracket this but do not replace logged batches.
- The baffle insert's heat transfer gain (x3) and pressure loss (4 velocity heads) are assumptions; fouling will reduce the gain over a season.
- Surface coefficients assume still air. Wind raises the shell loss and the wood demand.
- The methane figure is a literature average, not a prediction for this design.
- Nothing here has been measured. Checking these numbers against logged batches is TRL 4 work, which is on hold by Amish's instruction.

Safety: These are paper estimates for a fire that produces flammable, toxic gas and hot water. The surface temperatures in Table 3 (120 to 257 °C on the shell, higher on the flue) burn skin on contact. With the baffle and jacket blanket a long batch brings the water within about 1.5 MJ of boiling, so the water jacket must stay open-vented and at least three-quarters full. See CCB-PRC-001, Safety.